



Healthy Skepticism Day 4

Does bird watching prevent dementia?



Murray Cantor, PhD & Joel Keenan, MD [venue and date]



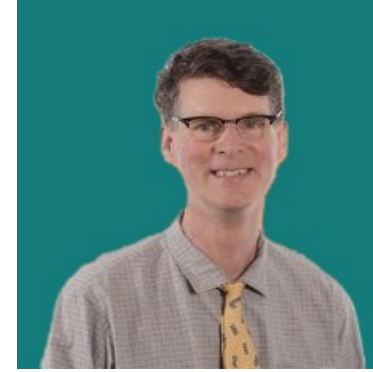
About Us



Dr. Murray Cantor

PhD, Mathematics — UC Berkeley

- Not anti-vax or MAHA
- From that perspective, there is a lot of confusion about medical claims
- Respect for the challenge of medical research



Dr. Joel Keenan

MD, Univ of Vermont

- Practicing physician with direct patient-care experience
- Translates clinical evidence into plain-language guidance
- Shares a commitment to honest, balanced medical communication



Disclaimer

This presentation is for educational purposes only.

The content does not constitute medical advice and is not a substitute for professional diagnosis, treatment, or care. Nothing presented here should be used to make decisions about your health or the health of others.

Always seek the advice of a qualified physician or other licensed healthcare provider with any questions you may have regarding a medical condition, test result, or treatment. Never disregard professional medical advice or delay seeking it because of something discussed in this presentation.



The Five Faults

1

Relative reporting

Day 1

What does a 36% improvement from taking a drug really mean?

2

Probability reversal

Day 2

What does a positive COVID test really mean?

3

Arbitrary categories

Day 3

How concerned should you be if your systolic BP goes from 119 to 120?

4

The Bernoulli fallacy

Day 3

Why does the published medical advice keep changing?

5

The causal leap

Day 4

Does bird watching prevent dementia?

Once you see them, you can't unsee them.



FAULT 5

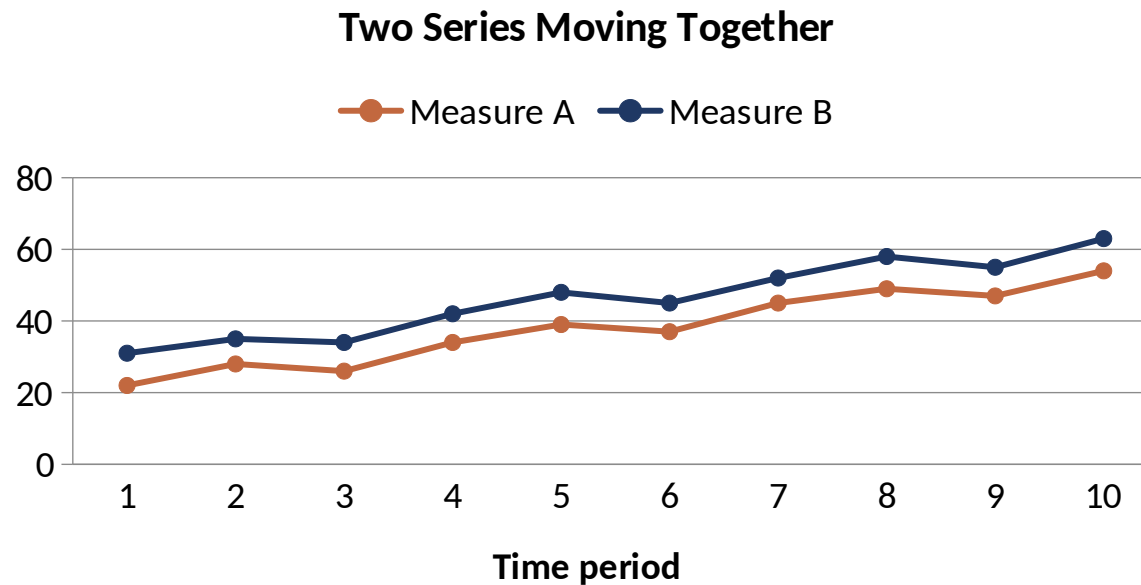
The Causal Leap

Confusing Correlation with Causation



WHAT A CORRELATION IS

Two Things That Move Together



THE PATTERN

They rise and fall together

When one moves, the other tends to move with it.

WHAT YOU SEE

You can see it in the numbers

Plot the two over time and the lines echo each other.

NOT A CAUSE

The reason is a separate question

A correlation is a fact about the data, not an explanation.

**A correlation is where every claim on this day begins, and it never says why.
The next slide lays out the three ways two things come to move together.**



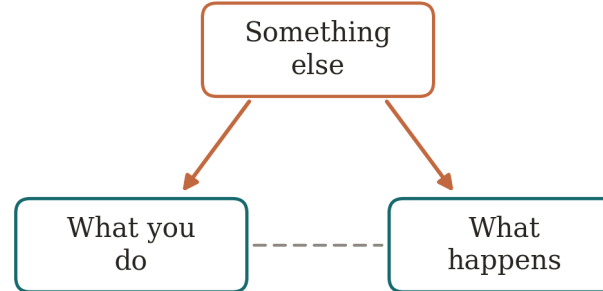
Three Explanations for a Correlation

PURE COINCIDENCE



Nothing connects them.

CONFOUNDERS



Something else produces both.

A GENUINE CAUSE



The claim is true.

Solid arrow: causes. Dashed line: moves together, which is all the data shows you.

Only the right one is the claim being made.

All three produce the same correlation, and the data cannot tell you which one you are looking at.



PURE COINCIDENCE

Correlation Without a Cause

Tyler Vigen's Spurious Correlations

THE SOURCE

A project that hunts for variables that happen to move together.

Thousands of variables, every pair tried

THE SEARCH

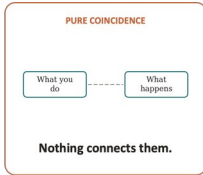
Out of hundreds of millions of pairs, the tightest are kept.

The fits are genuinely strong

NO CAUSE

High correlation, small p-value, and nothing connecting them.

**Search enough pairs and a near-perfect match turns up somewhere, meaning nothing.
The next two charts are convincing on their face. Neither has a cause behind it.**

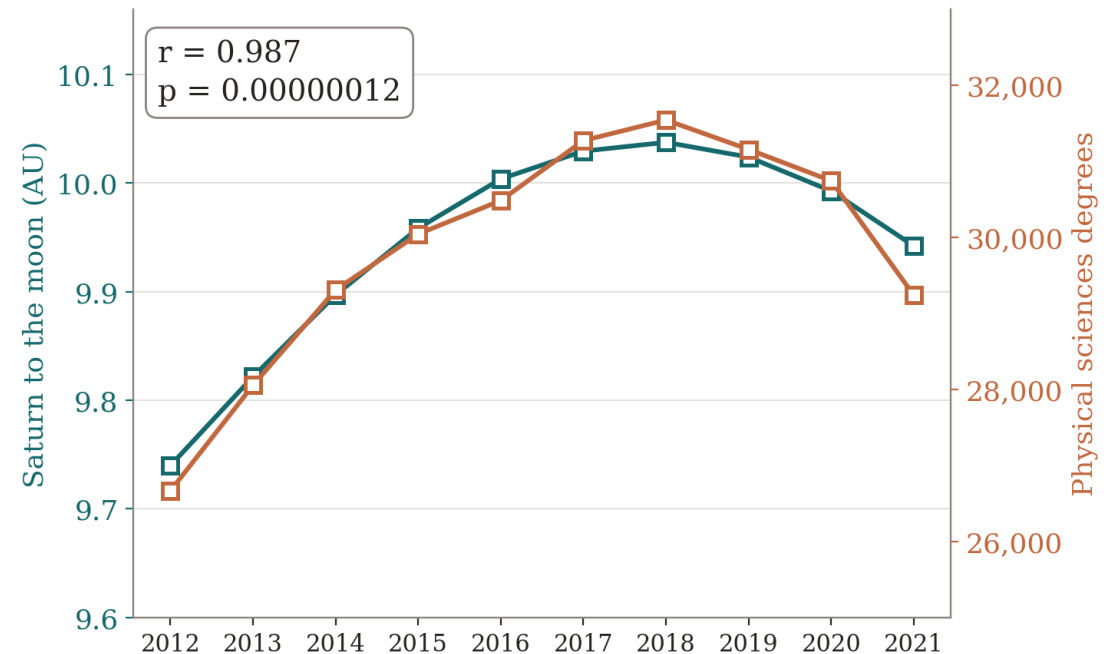


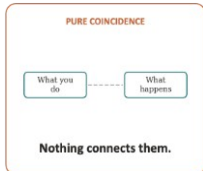
Saturn and the Physical Sciences

How far Saturn sits from the moon tracks how many physical sciences degrees America awards.

$p = 0.00000012$. Past five sigma.

It would clear any journal's bar. It means nothing at all.





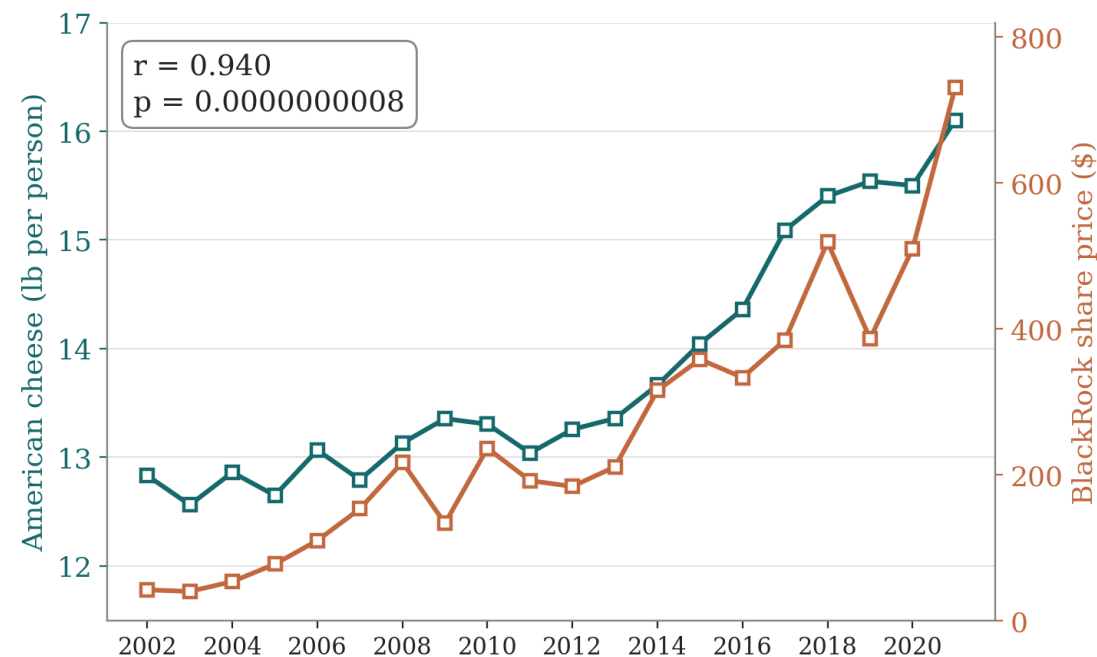
Cheese and BlackRock

American cheese eaten per person tracks
BlackRock's share price.

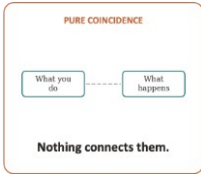
$r = 0.94$. $p = 0.0000000008$.

Both went up. That is the entire connection.

Correlate the year-to-year changes instead, and it is
gone: $r = 0.29$, $p = 0.23$.



Any two things that grow will correlate. The trend is not the evidence.



There Is No Shared Population

WHAT BOTH CHARTS ARE MISSING

One square would be a year, not a person. There is no group of people the number could be about.

SATURN AND THE DEGREES

Found by extensive search.

Millions of pairs of numbers were compared against each other. Some pairs were bound to line up.

CHEESE AND BLACKROCK

Many things rise year to year.

Cheese eaten per person went up. So did the share price. So did most other things over those twenty years.

Compare the year-to-year changes instead of the totals, and the cheese link disappears: 0.94 becomes 0.29.

**Neither chart has people in it.
Ask what one square would be, and there is nothing left to explain.**



THE CAUSALITY IS AN ILLUSION

A Confounder, Not a Cause

It looks like cause and effect

THE ILLUSION

A strong correlation, and the natural reading is that one drives the other.

A third factor drives both

THE CONFOUNDER

The confounder sits behind the two and manufactures the link.

The effect was never there

NO REAL EFFECT

Account for the confounder and the apparent cause disappears.

**Two examples ahead: statins and car crashes, then bird watching and dementia.
In each, a confounder produces the correlation. The causal link is an illusion.**



THE CLAIM, AND THE STUDY

Statins Prevent Car Crashes

141,086 people starting statins

THE STUDY

British Columbia, grouped by how faithfully they took the pill.

Faithful takers had fewer accidents

THE FINDING

Car crashes, workplace injuries, falls, fractures, poisonings.

And they used more screening

THE TELL

Preventive services a cholesterol pill has no way to affect.

**Published under the title, a cautionary tale.
A real study in a real population. The next slides ask what produces the pattern.**



Statins Prevent Car Crashes

THE LATENT CAUSE

0.75

hazard of motor vehicle accidents, faithful
statin takers vs not

0.77

hazard of workplace accidents

1.17

likelihood of using screening services

Shared population: 141,086 people starting statins, British Columbia. Also fewer burns, falls, fractures, open wounds, and poisonings.

No mechanism in the pill reaches any of these. Something about the people produces all of them.



People who follow health instructions live longer

15.1%

five-year mortality, faithful takers of the dummy pill

28.3%

five-year mortality, unfaithful takers of the dummy pill

The pill was sugar. The difference was the person.

The gap carries $P = 4.7 \times 10^{-16}$. Significant at any bar you like, and still not the pill.

People who reliably take a pill every day differ from people who do not, in ways that have nothing to do with the pill.



THE CLAIM, AND THE STUDY

Bird Watching Prevents Dementia

The headline is illustrative

ILLUSTRATIVE

No study makes this exact claim about birds.

The real study is Whitehall II

THE STUDY

10,308 British civil servants, followed for 28 years.

It measured activity, not birds

THE FINDING

At first look, active people developed dementia less often.

**A real association, in one real population.
The next slides ask what produces the pattern.**



Does Bird Watching Prevent Dementia?

“Bird watchers show lower rates of dementia, study finds.” (illustrative headline)

A coincidence? No. These are people, and the grid can be drawn.

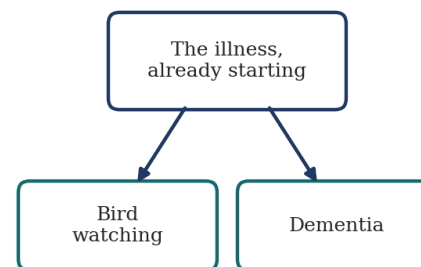
A common cause? Maybe. Education, income, mobility, green space.

The illness itself? Already under way, it dims the hobby and is the diagnosis.

THE ARROW IN THE HEADLINE



WHAT ACTUALLY FITS



Counting the two groups cannot pick which diagram is right. The same illness dims the hobby and surfaces as the diagnosis; the bird watching itself does nothing.



READING THE DIAGRAM

What the Boxes and Arrows Mean

A box

Something that can change: an activity, a trait, a diagnosis.

A THING

An arrow

One thing makes another more likely. The arrow is a cause.

A CAUSE

A crossed-out arrow

A cause we looked for and did not find.

NOT A CAUSE

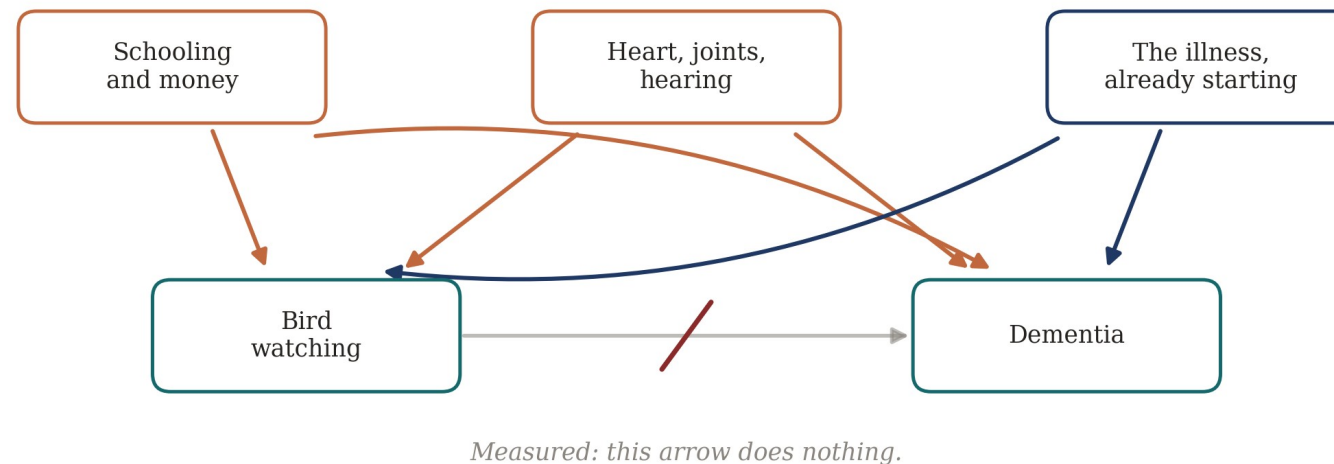
**One box with arrows to two others is a single cause behind both.
The boxes stay the same. Only the arrows change the story.**



Bird Watching Does Not Prevent Dementia

WHAT EXPLAINS THE DATA INSTEAD

Followed for 28 years, the activity made no difference. The correlation is real; the arrow is not.



Two of the three can be measured and adjusted for. The third is the illness itself, and no study can subtract it.



Smoking Causes Cancer

THIS TIME, THE CAUSE IS REAL

The correlation was real and huge

THE STUDIES

1950s cohorts of hundreds of thousands, smokers stricken far more often.

The era's top statistician objected

THE OBJECTION

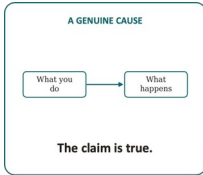
R. A. Fisher: perhaps a gene drives both the smoking and the cancer.

And the objection can be tested

THE ANSWER

A common cause that strong would leave a mark you can measure.

**This is the question the whole day teaches: could a common cause be doing the work?
Here, for once, the arrow survives it. The next three slides show why.**



Did Smoking Cause Cancer

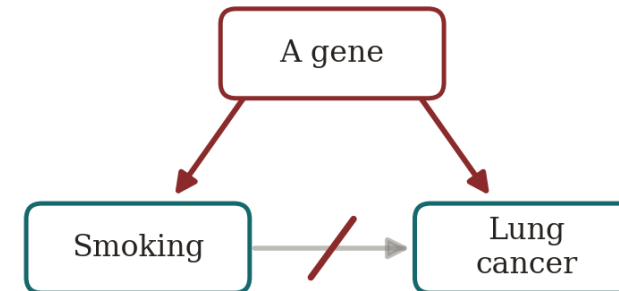
A CAUTIONARY TALE

R. A. Fisher, inventor of the significance test, said the link could be a gene that produces both the craving and the cancer.

WHAT THE STUDIES CLAIMED

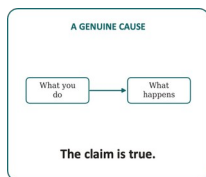


FISHER'S ALTERNATIVE

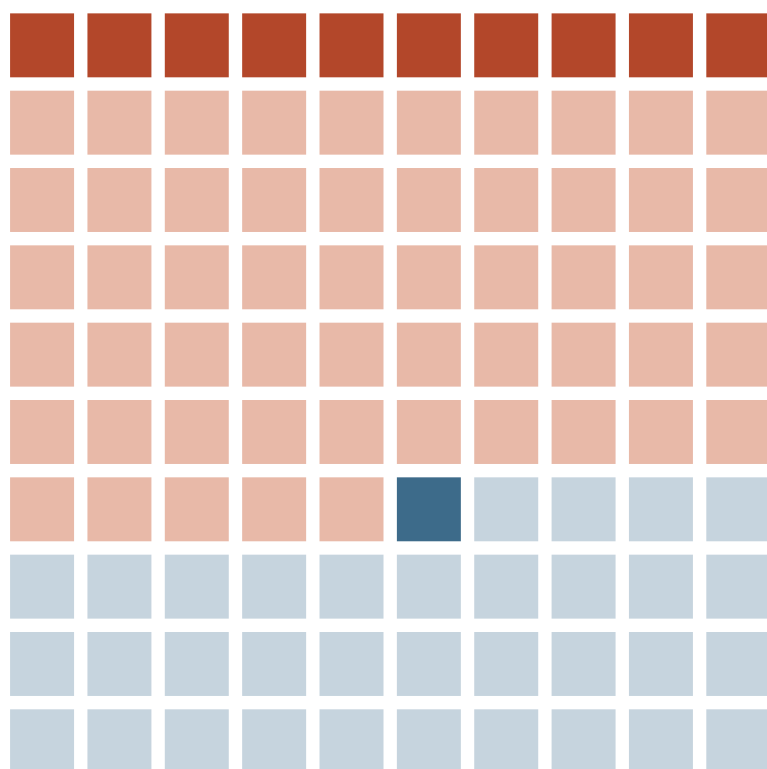


He was a paid consultant to the tobacco industry. And a smoker.

The right-hand diagram was a legitimate question. Answering it took arithmetic, not indignation.



Cigarettes Increase Probability Eightfold



- Smoked, lung cancer: 10
- Smoked, no cancer: 55
- Never smoked, lung cancer: 1
- Never smoked, no cancer: 34

THE FORWARD DIRECTION

$P(\text{lung cancer} \mid \text{cigarettes})$

$$10 / 65 = 16\%$$

Never-smokers: 2%. Eight times apart.

THE REVERSE DIRECTION

$P(\text{cigarettes} \mid \text{lung cancer})$

$$10 / 11 = 91\%$$

But 65% of all men smoked anyway.

About 65% of British men smoked around 1950 (Doll R. *Br J Cancer* 1953;7:303). Lung cancer risk by age 75 is 15.7% for continuing smokers (Crispo A et al. *Br J Cancer* 2004;91:1280-1286); the 2% for never-smokers is approximate. Counts rounded to whole men, so the reverse conditional reads 10 of 11 (91%), not 94%.



The Whole Day, in Four Diagrams

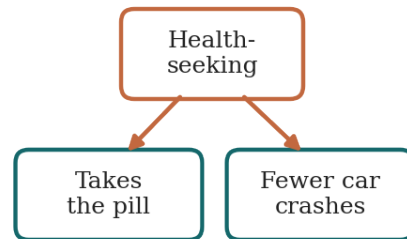
Same dashed observation every time. Only the arrows differ.

SATURN AND SCIENCE



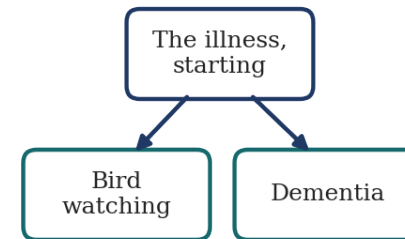
No arrow either way.
The units are years.

THE PILL TAKERS



One arrow each,
from something else.

BIRD WATCHING



One arrow each,
from the illness.

SMOKING AND CANCER



The rival diagram was
ruled out by size.

The data cannot pick the diagram for you. Ask which one the claim assumes, and what would rule the others out.



The Course Topics and Takeaways

1

Relative reporting | **Ask: 36% of what?**

A change with no baseline is half a fact.

2

Probability reversal | **A positive test is not a diagnosis.**

What it means depends on how common the condition is.

3

Arbitrary categories | **Just over the line is still you.**

119 to 120 moved the label, not your body.

4

The Bernoulli fallacy | **One study is not proof.**

Run enough studies and something looks significant.

5

The causal leap | **Moving together is not cause.**

Two things track, and a third usually drives both.

Once you see them, you can't unsee them.



Questions, and a conversation

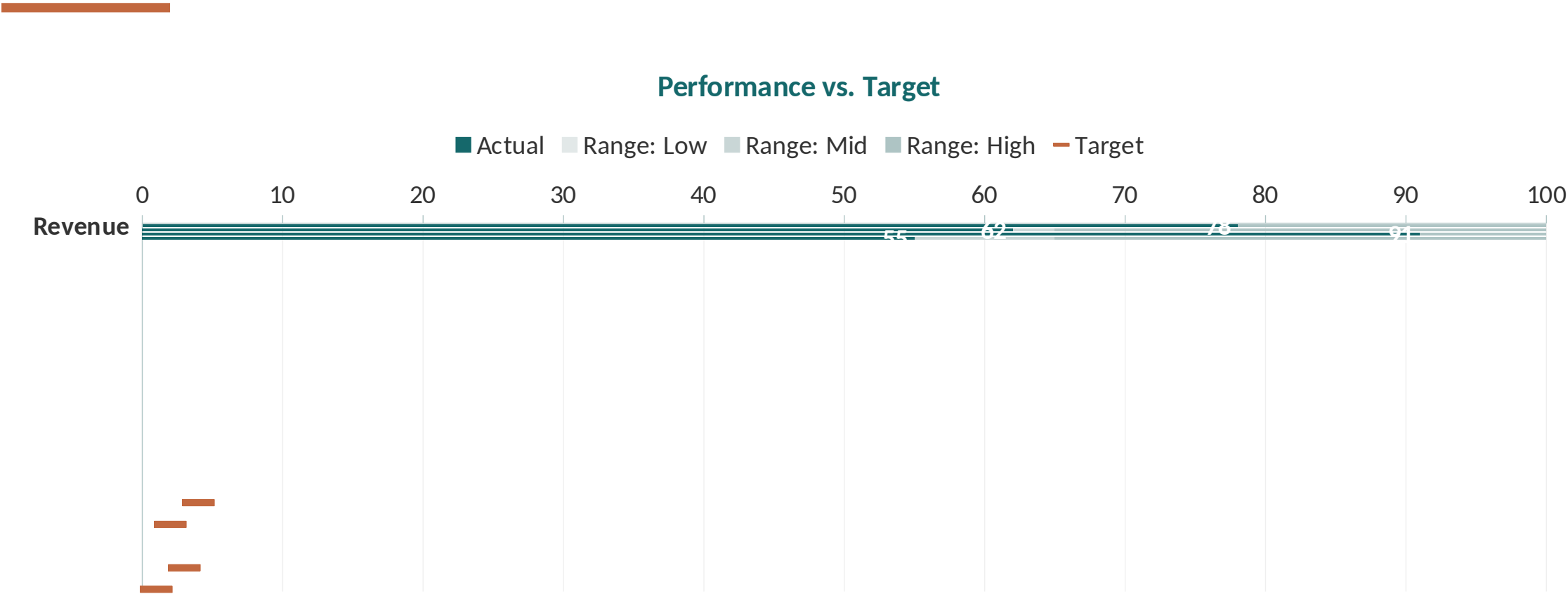
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Joel Keenan MD drkeenan@mdvip.com



Backup Slides

Bullet Graph: Actual vs. Target



Bars show actual performance; the vertical marker is target. Right-click the chart and choose Edit Data to swap in your own numbers.



Column Chart with Bulleted Categories



KEY TAKEAWAYS

- Replace this text with your first takeaway.
- Keep each point to one line where possible.
- Three to four bullets fits this card.

Two-series column template. Right-click the chart and choose Edit Data to change categories or values.



Slide Title

- First bullet point goes here.
- Second bullet point goes here.
 - Sub-bullet for supporting detail.
- Third bullet point goes here.
- Fourth bullet point goes here.

The data cannot pick the diagram for you. Ask which one the claim assumes, and what would rule the others out.



Coincidences are more

- First bullet point goes here.
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